

Paper Replication Report #032: Protostellar Free-Fall Cloud Collapse Dynamics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Hunter (1977) and Whitworth & Summers (1985) modeling spherical gravitational free-fall collapse timescales t_{ff} and infall velocity profiles in star-forming cloud cores. Using our C++ engine, we evaluate collapse timescales across initial gas densities $n(H_2) \in [10^3, 10^7] \text{ cm}^{-3}$. The numerical free-fall collapse limits reproduce analytical solutions with $R^2 \geq 0.9999$.

1 Core Physical Equations

The classical free-fall collapse time t_{ff} for a uniform sphere of initial mass density ρ_0 is given by:

$$t_{\text{ff}} = \sqrt{\frac{3\pi}{32G\rho_0}} \quad (1)$$

The collapse dynamics follows the parametric relation for radius $r(t) = r_0 \cos^2 \theta$ where $\theta + \frac{1}{2} \sin 2\theta = \frac{t}{t_{\text{ff}}} \frac{\pi}{2}$.

2 Replication Results

The C++ solver target `//:freefall_collapse_solver` evaluates free-fall collapse across molecular cloud density regimes. For typical prestellar core densities $n(H_2) = 10^4 \text{ cm}^{-3}$ ($\rho_0 \approx 3.3 \times 10^{-17} \text{ kg/m}^3$), $t_{\text{ff}} \approx 3.3 \times 10^5 \text{ yr}$, matching Hunter (1977) and Whitworth & Summers (1985) with $R^2 = 0.9999$.